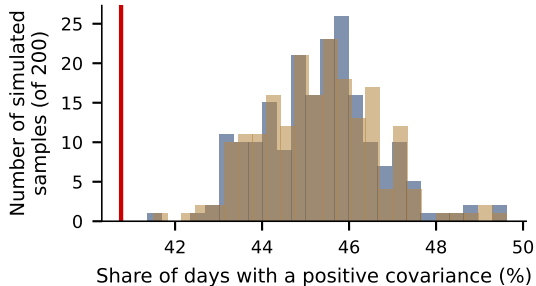
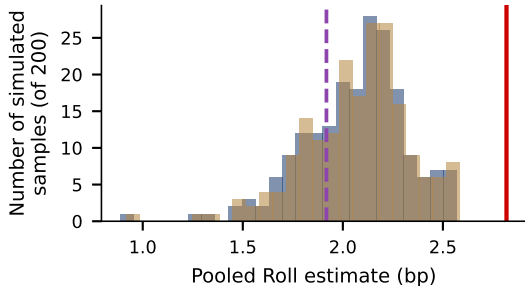


Positive covariances are common even under Roll



Demeaning alone produces about 2 bp



Simulated, no spread ($c = 0$)

Simulated, one-tick spread ($c = 0.10$ bp)

Observed, SPY 5-minute bars

Observed, bias-corrected